Magic System & Mathematical Integration — Version 0.10.01

Established canon preserved
Working model
Open design decisions

Document scope: the relationship between mathematics, sigils, spell construction, Magic Sprites, magical interfaces, and gameplay-based learning. This is a focused semi-story bible, not a complete rewrite of the MATHBOUND universe.

0. Purpose of this document

MATHBOUND needs a magic system that can accommodate the full range of school mathematics, beginning with basic competencies such as counting and progressing toward increasingly abstract concepts.

The system must also preserve the project's fantasy identity. Mathematics should not become a collection of arbitrary magical powers, and spellcasting should not become a conventional quiz in which players answer a question to unlock an attack.

This version consolidates the current direction into one working framework. It preserves established principles from the earlier story bible while distinguishing confirmed canon from proposed mechanics and decisions that remain open.

The central design principle is:

> Mathematics provides the formal structure through which magic is understood and expressed. Sigils are used to construct spells. Magic Sprites interpret and enact those expressions within the limits of their nature and the world. Players discover how to use this relationship through mathematical understanding.

The goal is not to make mathematics magical simply by attaching fantasy names to it. The goal is to make mathematical understanding an integral part of how players interact with, investigate, and influence the world.

I. Foundational principles

1. Mathematics does not power magic

Status: Established canon

Mathematics is not an energy source, a fuel, or an independent magical force.

Magic already exists within the world. It manifests through the environment and through Magic Sprites, which possess their own natures and capabilities. Mathematics provides a way to understand, represent, and communicate with aspects of that reality.

This distinction is essential.

If mathematics itself were the fuel for magic, the game could easily become a system in which players accumulate mathematical knowledge as magical energy. That is not the intended direction.

Instead, mathematical knowledge allows players to express more precise intentions, understand magical phenomena, and construct more sophisticated interactions.

Example:

A player encounters a stone floating above a pond.

- The stone's floating behavior is a magical phenomenon.
- A nearby Magic Sprite is responsible for, or participates in, the phenomenon.
- The player observes the stone and investigates its behavior.
- Mathematical concepts may help the player describe, predict, or influence the phenomenon.
- A correctly constructed magical expression can communicate an intended interaction, provided the Sprite and circumstances permit it.

The mathematics does not generate the phenomenon from nothing. It helps the player understand and interact with what exists.

2. Reality constrains magical intention

Status: Established canon

The player may envision a desired outcome, but imagining something does not guarantee that reality will comply.

A magical expression must be meaningful within the established system. The Sprite must be capable of interpreting and enacting it, and the circumstances must permit the intended effect.

Consequently, a mathematically valid expression does not automatically guarantee a successful magical outcome.

There are at least three distinct questions:

1. Is the mathematical expression correct?

2. Can the Sprite interpret and execute the requested interaction?
3. Can the intended effect occur under the present conditions?

These questions must remain distinguishable in both the lore and gameplay.

Example:

A player correctly constructs an expression intended to move a large stone. The expression is mathematically valid, but the available Sprite cannot influence objects of that kind or magnitude.

The spell may fail to produce the intended movement. That failure does not mean the mathematics was incorrect.

This principle protects the educational integrity of the system while allowing magical limitations to remain meaningful.

3. Discovery is the primary relationship between mathematics and magic

Status: Established canon, with proposed gameplay implications

MATHBOUND's central theme is discovery: there is always more to discover.

Players should not simply receive a list of magical effects and learn the answers required to activate them. They should investigate the world, encounter unfamiliar phenomena, form interpretations, test their understanding, and discover more effective ways of interacting with their surroundings.

Mathematics is one of the means through which this discovery occurs.

For example, counting can reveal the number of objects in a collection. Comparing quantities can reveal a discrepancy. Recognizing relationships can help a player predict an outcome. More advanced mathematics can support more sophisticated investigations and constructions.

These activities should feel like parts of an RPG adventure rather than isolated exercises inserted between story scenes.

II. The architecture of magic

4. The Two-Table system

Status: Established canon

The Two-Table system remains the foundational model for magical interaction.

TABLE I — ENVISION

The outcome the practitioner intends to achieve.

What do I want to happen?

Formal communication

TABLE II — COMMAND

The means by which the practitioner communicates the intended interaction.

How do I express what I want to happen?

Sprite interpretation

MAGICAL INTERPRETATION

The magical interaction is interpreted according to the Sprite's nature, understanding, and capabilities.

RESPONSE

The resulting magical phenomenon, including success, partial execution, an unexpected effect, or failure.

The distinction between Envision and Command must remain intact.

Envision is the intended outcome. Command is how that intention is formally communicated. The two are related, but they are not interchangeable.

The revised sigil-based concept belongs primarily within the Command side of the system: sigils give the practitioner a structured way to express an intended interaction.

Example: Opening a magical gate

A player wants to open a gate.

- Envision: The player intends for the gate to open.
- Command: The player constructs an appropriate magical expression using the available sigils.
- Magical Interpretation: The relevant Sprite interprets the expression and determines whether it can enact the intended interaction.
- Response: The gate opens, remains closed, responds incompletely, or produces another outcome consistent with the system's rules.

This example illustrates the architecture rather than establishing a finalized gate-opening spell.

The player should not be required to understand the gate's entire magical mechanism before interacting with it. However, learning more about the gate and its Sprite may reveal additional ways to interact with it.

5. Mathematical language, magical phenomena, and interfaces

These are related components, but they must not be treated as the same thing.

Component	Role
Mathematics	Provides precise concepts, relationships, and structures that can be understood and represented.
Sigils	Represent specifically defined mathematical concepts, operations, or properties.
Mathematical expressions	Arrange mathematical elements into meaningful structures according to their rules.
Spells	Constructed magical instructions or interactions communicated through the magical system.
Magic Sprites	Living manifestations that interpret and enact magical interactions within their capabilities.
Magical devices	Interfaces that facilitate magical communication, interpretation, or execution.

Environment	Provides the conditions, phenomena, and limitations within which magic operates.

An important terminology distinction is retained here: sigils, mathematical expressions, and graphs are not interchangeable categories. Their exact relationships and possible representations remain a matter for the formal language design.

A spell may involve multiple mathematical elements, but the exact rules governing which elements are sigils, which are inputs, and how they form a spell have not yet been fully defined.

III. Sigils and mathematical representation

6. What is a sigil?

Status: Established principle; detailed implementation open

A sigil is a magical representation corresponding to a specifically defined mathematical concept, operation, or property.

Its meaning is not arbitrary. A sigil cannot mean one mathematical thing in one situation and an unrelated thing in another simply because the story needs it to.

For example, if a sigil represents addition, its mathematical meaning must remain addition. Its magical applications may vary, but the underlying mathematical meaning does not.

This is one of the most important constraints in MATHBOUND.

6.1 One mathematical meaning, potentially many applications

Consider addition.

Addition could be used in different encounters to:

- Determine a combined quantity of objects.
- Calculate the total resources needed for a task.
- Combine quantities associated with a magical construction.
- Express part of a more complicated spell.

These applications differ, but they do not require the addition sigil to acquire a different mathematical meaning each time.

The same principle should apply to other mathematical concepts as they enter the game.

6.2 Not every value needs its own sigil

A mathematical expression can contain both structural elements and values used within that structure.

For example, the numeral 5 may serve as an input representing a quantity in a counting-related spell. That does not automatically mean the number five must be a separate magical sigil.

The number's representation, the concept of counting, and the structure of the resulting expression are different design questions.

We should define them precisely rather than assume they all require distinct symbols.

6.3 Not every mathematical competency requires a new magical property

A competency may introduce a new concept, develop an existing concept, or teach students to use a concept in a different context.

The magic system should accommodate these differences.

If students learn to count larger quantities, for example, the game does not necessarily need to introduce a new magical element. The existing mathematical framework may be sufficient, with the difficulty arising from the quantity, representation, context, or task.

Likewise, a more advanced competency might require a new mathematical structure without requiring an entirely new type of Sprite.

Design rule: New mathematical learning should expand what players can understand and express, not automatically expand the world's catalogue of magical powers.

7. How sigils become spells

Status: Working model

The current direction is that sigils are used to construct spells.

A simple spell may require only one sigil or a simple expression. More sophisticated spells may require multiple sigils arranged according to the rules of the mathematical language.

The progression is therefore not merely about collecting more symbols. It is about learning how to use mathematical structures.

Level 1

Basic expression

A simple mathematical structure supports a limited, well-defined interaction.

One defined mathematical element

Example: A counting-related expression used to identify the quantity of a collection.

Level 2

Composite expression

Multiple mathematical elements form a valid, structured expression.

Illustrative structure: two quantities combined

Example: A spell that works with two collections and their combined quantity, where the encounter and Sprite support that interaction.

Level 3

Advanced expression

More sophisticated structures express relationships, constraints, or coordinated operations.

Multiple related mathematical structures

Example: A constructed spell that must account for several related quantities or conditions before a magical mechanism can respond as intended.

These levels are explanatory examples, not a finalized spell-level or tier system.

7.1 Complexity must be meaningful

More sigils should not automatically mean greater power.

A longer expression might enable the player to:

- Handle additional inputs.
- Represent a more complicated relationship.
- Coordinate several operations.
- Express more precise conditions.
- Control a more complicated interaction.

The actual outcome still depends on the mathematical meaning of the expression and the Sprite's capabilities.

An unnecessarily long expression should not become more powerful merely because it contains more symbols.

7.2 Spell grammar remains an open decision

For the system to be consistent, it eventually needs rules for:

- Which mathematical elements may be combined.
- How those elements must be ordered or grouped.
- Which combinations form valid expressions.
- How values and other inputs enter an expression.
- How the resulting expression communicates an intended interaction.
- How an invalid expression differs from a valid expression that a Sprite cannot execute.

We should not invent all these rules at once. The counting competency will serve as the first practical test of the system.

IV. Magic Sprites

8. Nature of Magic Sprites

Status: Established canon

Magic Sprites are living, personality-bearing manifestations of magic associated with particular environments and phenomena.

They are not simply batteries, passive tools, or interchangeable magical energy sources.

Their natures influence what they can do, how they behave, and how they respond to practitioners.

Some aspects of their behavior may be understood through observation and mathematical investigation. Other aspects may remain mysterious, preserving the fantasy of the world.

This creates a deliberate distinction between understanding a phenomenon and completely explaining its underlying mechanism.

9. Three levels of understanding a Sprite

A useful working model distinguishes three levels of knowledge.

LEVEL 1 — OBSERVATION

What happens?

The player notices a Sprite's behavior and the effects it produces.

LEVEL 2 — MATHEMATICAL MODEL

What relationships can be understood or predicted?

The player uses relevant mathematical concepts to describe patterns, quantities, or relationships in the observed behavior.

LEVEL 3 — UNDERLYING MECHANISM

Why does the phenomenon exist in the first place?

The Sprite's fundamental supernatural nature may remain partly or entirely unexplained.

Example: The floating-stone Sprite

A Sprite causes several stones to float above a pond.

- The player observes that the stones are floating.
- The player counts them, compares their movement, or records how their positions change.
- Mathematical reasoning helps the player predict or describe some of their behavior.
- The underlying reason that the Sprite can suspend stones in the air may remain a magical mystery.

The player does not need to solve the entire mystery of magic to learn something meaningful about the phenomenon.

This is important for MATHBOUND's discovery theme: understanding can grow without requiring every mystery to be completely resolved.

10. Sprite affinity, environment, and distance

Status: Established principle; detailed rules open

Sprites are associated with the places and phenomena from which they emerge. Their behavior may be influenced by environmental conditions and their relationship to their surroundings.

A Sprite associated with fire, for example, might behave differently in a cold environment than in an environment suited to its nature.

Distance from its associated environment may also matter, depending on the Sprite.

However, these rules should not become universal assumptions that every Sprite follows identically. Environmental affinity, distance, and individual capabilities are distinct factors.

Example

A player constructs a valid expression intended to move a collection of floating stones.

The expression is correct, and the Sprite understands the requested interaction. However, the Sprite's abilities are weakened by the present environmental conditions.

The outcome may be incomplete or less effective.

The important distinction is that the mathematical expression did not become incorrect. The circumstances changed the Sprite's ability to enact it.

11. Understanding and befriending Sprites

Status: Established direction; exact mechanics open

Understanding a Sprite and developing a relationship with it may improve the reliability or stability of magical interactions.

This offers a way to connect the RPG's social and exploration systems with its magic system.

Players could discover a Sprite's habits, preferences, environmental needs, and capabilities. This knowledge may reveal how to communicate with it more effectively.

However, befriending a Sprite should not replace mathematical understanding.

A strong relationship might help a Sprite interpret or enact a valid expression, but it should not turn a mathematically incorrect expression into a mathematically correct one.

Example:

A player has learned to construct an appropriate counting-related spell but initially struggles to communicate it effectively to an unfamiliar Sprite. Through exploration and interaction, the player learns how the Sprite responds to different forms of communication.

The improved relationship may make the interaction more reliable. The mathematical meaning of the counting-related expression remains unchanged.

V. Magical devices and spell execution

12. Magical devices are interfaces, not batteries

Status: Established canon

Magical devices facilitate interaction with magic. They do not function as personal reservoirs of magical energy.

Their role is related to communication, interpretation, and execution.

A more advanced device might allow a practitioner to communicate more reliably, construct more complicated expressions, or interact with a wider range of magical phenomena. The exact capabilities depend on the device's eventual design.

A device should not automatically grant mathematical understanding to its user.

Example:

Two players attempt to communicate the same valid expression to a compatible Sprite.

One uses a rudimentary interface and struggles to express the structure reliably. The other uses a more capable device that supports more precise communication.

The second player may have an execution advantage, but the device does not mean that the player understands the mathematics better.

This distinction prevents equipment progression from replacing educational progression.

13. Factors affecting spell outcomes

The following framework consolidates the factors discussed so far.

Factor	Primary question
Mathematical correctness	Is the expression mathematically valid and appropriate for its intended meaning?
Expression construction	Have the mathematical elements been arranged according to the system's rules?

Sprite capability	Can the Sprite perform the intended interaction?
Sprite understanding and relationship	Can the Sprite interpret and respond to the practitioner effectively?
Magical interface	Can the device communicate the expression reliably?
Environmental conditions	Do the current circumstances support the intended effect?
Complexity	Does the expression demand capabilities beyond those available?
Intended outcome	Is the requested interaction consistent with the actual structure of the expression?

These factors are conceptually distinct. They do not need to become eight independent numerical statistics in the game.

Some may be represented through explicit mechanics; others may be narrative constraints, discoverable rules, or encounter conditions.

The key requirement is that the game remains consistent about why a spell works or fails.

14. Types of spell failure

A useful diagnostic framework is:

Mathematical or structural failure

The expression is incorrect, incomplete, or arranged in a way that violates its formal rules.

Capability failure

The expression is valid, but the Sprite cannot perform the requested interaction.

Communication failure

The expression cannot be conveyed or interpreted reliably through the current interface or interaction.

Environmental limitation

The Sprite's capabilities or the intended phenomenon are affected by the surrounding conditions.

The player should receive enough feedback to distinguish these cases.

A failed spell should not always be presented as a generic “wrong answer” message. When appropriate, it should help the player discover whether the problem lies in the mathematics, the expression, the Sprite, or the circumstances.

VI. Integrating mathematics into actual gameplay

15. Mathematics should be useful, not merely required

Status: Core design direction

A central risk is turning spellcasting into a repetitive loop:

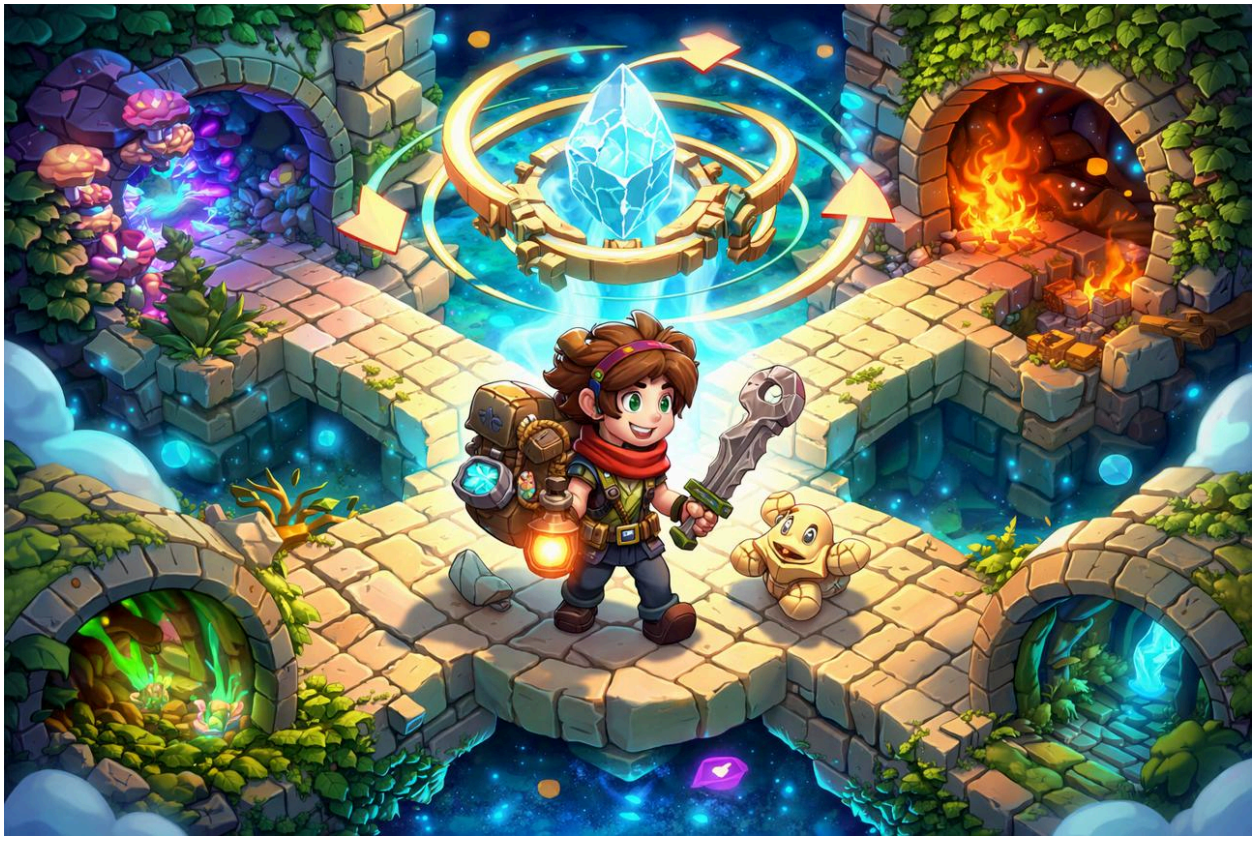
> See question → enter answer → cast spell → continue.

That sequence can introduce a concept, but it does not provide enough variety to carry an entire educational RPG.

Mathematics should influence the player's decisions and interactions with the world. Depending on the competency, a player might need to count, classify, compare, represent, estimate, calculate, recognize relationships, or construct a formal expression.

The relevant mathematical activity should be part of solving the encounter, rather than a separate gate placed in front of it.

Five ways mathematics can contribute to gameplay



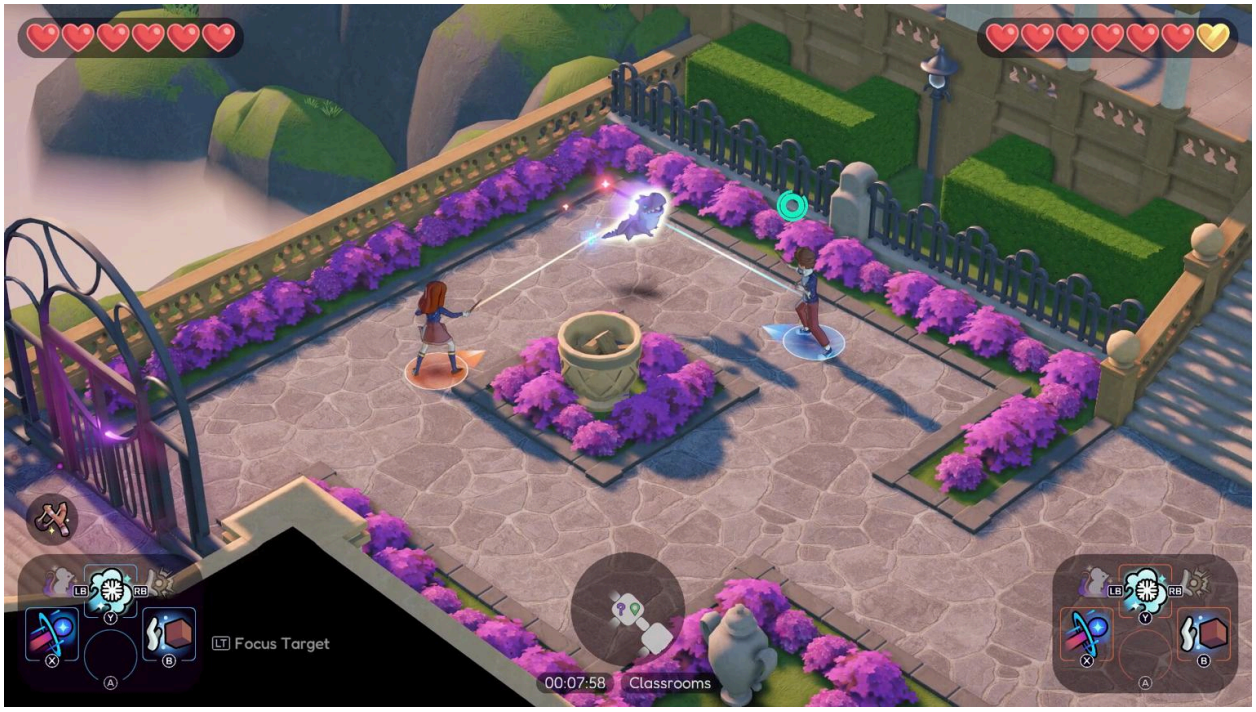
1. Exploration

Mathematical understanding helps the player investigate the environment, recognize quantities, and discover hidden information.



2. Puzzle solving

The player uses mathematical concepts to determine how a mechanism or phenomenon should be approached.



3. Combat

Mathematical mastery influences the effectiveness and sophistication of attacks, consistent with MATHBOUND's established combat direction.



4. Investigation and story

Mathematical reasoning helps the player discover discrepancies, interpret evidence, and understand events.



5. Interaction with Sprites

Players use mathematical understanding to interpret a Sprite's behavior and communicate more effectively with it.

These are different ways to apply the same educational framework. Not every competency needs to appear in all five.

16. Logical and intuitive approaches

Status: Established conceptual distinction; exact mechanics open

The system may support two broad approaches to understanding and interacting with magic.

- Formalized approach: The practitioner explicitly represents the intended interaction through mathematical structures and formal communication.
- Experiential approach: The practitioner begins with observation, experimentation, and interpretation of a phenomenon.

These approaches should not represent two unrelated kinds of reality. They are different ways of engaging with the same underlying magic.

An intuitive approach must still operate within the world's rules. It is not a means of bypassing mathematical truth whenever the player wishes.

Likewise, a formalized approach does not automatically reveal the complete underlying mechanism of a magical phenomenon.

Example: Understanding a Sprite's movement

A player observes a Sprite that moves between floating objects.

Through experimentation, the player discovers a pattern in its behavior. This is an experiential approach.

Later, the player develops a mathematical representation of that pattern and uses it to predict where the Sprite will move. This is a more formalized approach.

The second approach builds on the first. The player does not need to choose between a world of mathematics and a world of intuition.

The precise relationship between these approaches and the existing Two-Table system remains open for further design.

VII. Worked example: The first mathematical competency — Counting

This is the first practical test of the entire system. If we can make counting meaningful without inventing a new magical ability for every counting-related skill, we have a promising starting point.

17. Encounter concept: The Floating Stones



hyper3d.ai

Illustrative visual reference for the encounter; not established story canon.

Scenario

The player enters an academy courtyard where several stones float above a gap. A Magic Sprite inhabits the area and can influence the stones.

The player must cross the gap, but the stones need to be arranged or activated appropriately.

The encounter introduces the player to three ideas:

- Mathematical understanding can be applied to a concrete situation.
- Sigils provide a structured way to construct magical expressions.
- Magic Sprites interpret those expressions and enact interactions they are capable of performing.

Step 1 — Observe the phenomenon

The player sees the floating stones and investigates the situation.

Depending on the learning objective, the player may need to count the stones, determine how many are available, or identify how many are needed for the crossing.

The mathematical activity comes from the situation itself.

Step 2 — Envision the outcome

The player decides what needs to happen: for example, arranging the available stones to form a usable crossing.

The intended outcome belongs to the Envision side of the Two-Table system.

Step 3 — Construct the spell

The player uses the appropriate counting-related sigil and provides the necessary quantity in the form supported by the spell language.

For illustration, suppose the task requires the player to communicate that a collection contains five stones.

The number 5 is a value supplied within the interaction. It is not automatically a separate sigil.

The exact written structure, notation, and input method remain undecided.

Step 4 — The Sprite interprets the expression

The Sprite receives the constructed expression and interprets the intended interaction.

If the expression is valid and the Sprite can perform the requested action under the current conditions, it attempts to enact it.

If the expression is invalid, the Sprite cannot interpret it as intended. If the expression is valid but the Sprite lacks the necessary capability, the interaction may fail for a different reason.

Step 5 — Observe the result

The stones respond, and the player receives feedback about the outcome.

The player crosses the gap if the encounter's requirements have been met.

The encounter introduces counting through a meaningful action rather than through an isolated question.

18. Expanding the same competency

The counting sigil does not need to change every time the player encounters a different counting task.

Application A

Count a collection

The player determines the number of objects present and uses that quantity to complete a task.

Application B

Investigate a discrepancy

The player counts objects in two collections and compares the quantities to discover that something has changed.

Application C

Organize resources

The player counts available materials and determines whether there are enough to complete a construction.

Application D

Investigate a Sprite

The player counts repeated appearances or objects associated with a Sprite to identify a pattern relevant to the encounter.

These examples demonstrate a central advantage of the framework: one mathematical concept can support multiple forms of gameplay without requiring a different magical property for each activity.

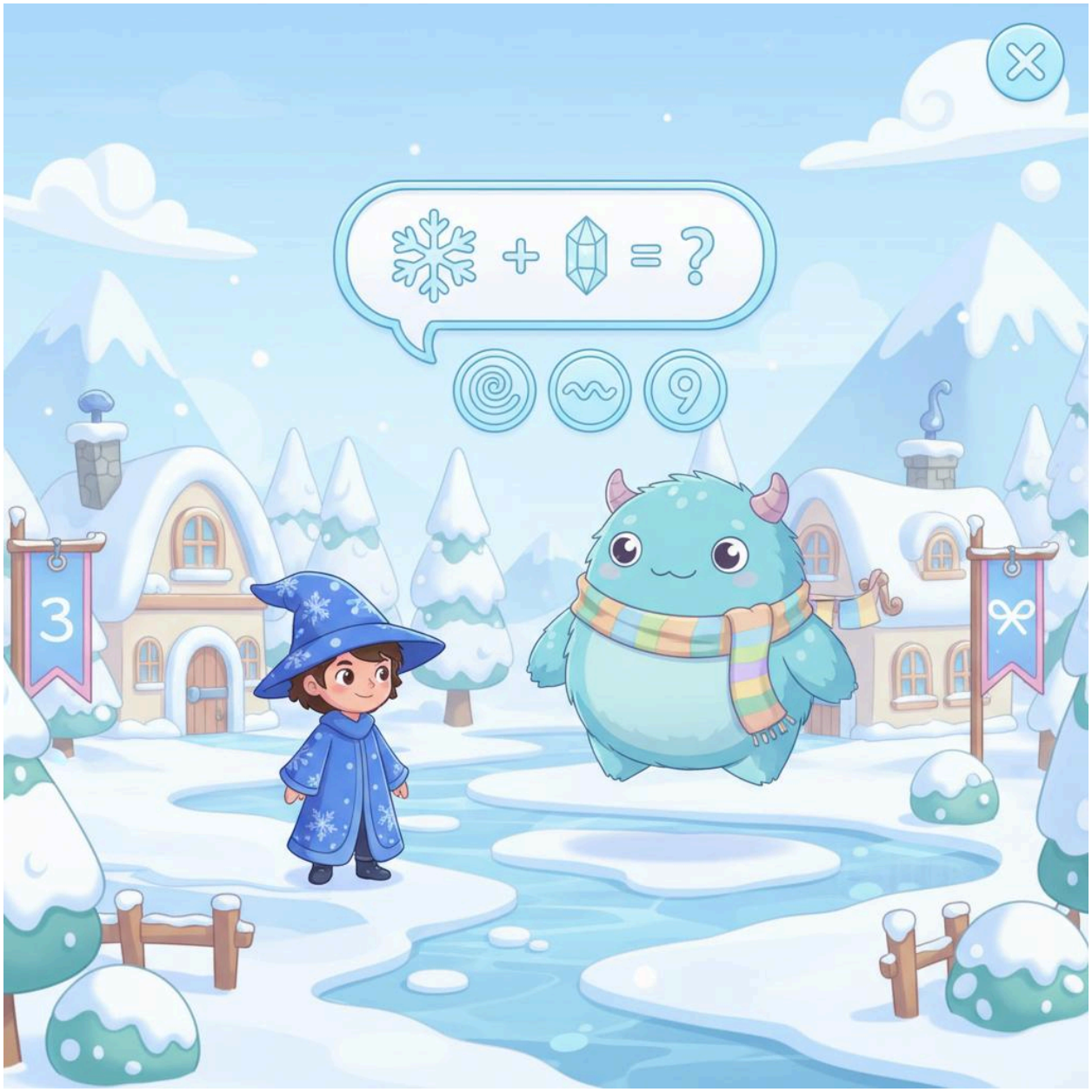
The actual encounters must still be mapped to the exact curriculum competencies. Counting objects, recognizing numerals, counting in sequence, and understanding cardinality are related but distinct learning objectives; they should not be treated as interchangeable merely because they involve numbers.

VIII. From counting to more advanced mathematics

19. The framework should scale with understanding

The counting encounter establishes the basic interaction. Later competencies can introduce more complex mathematical structures without changing the foundational architecture.

The following examples illustrate how the framework might grow.



Stage A — Counting

Mathematical focus: determining quantities.

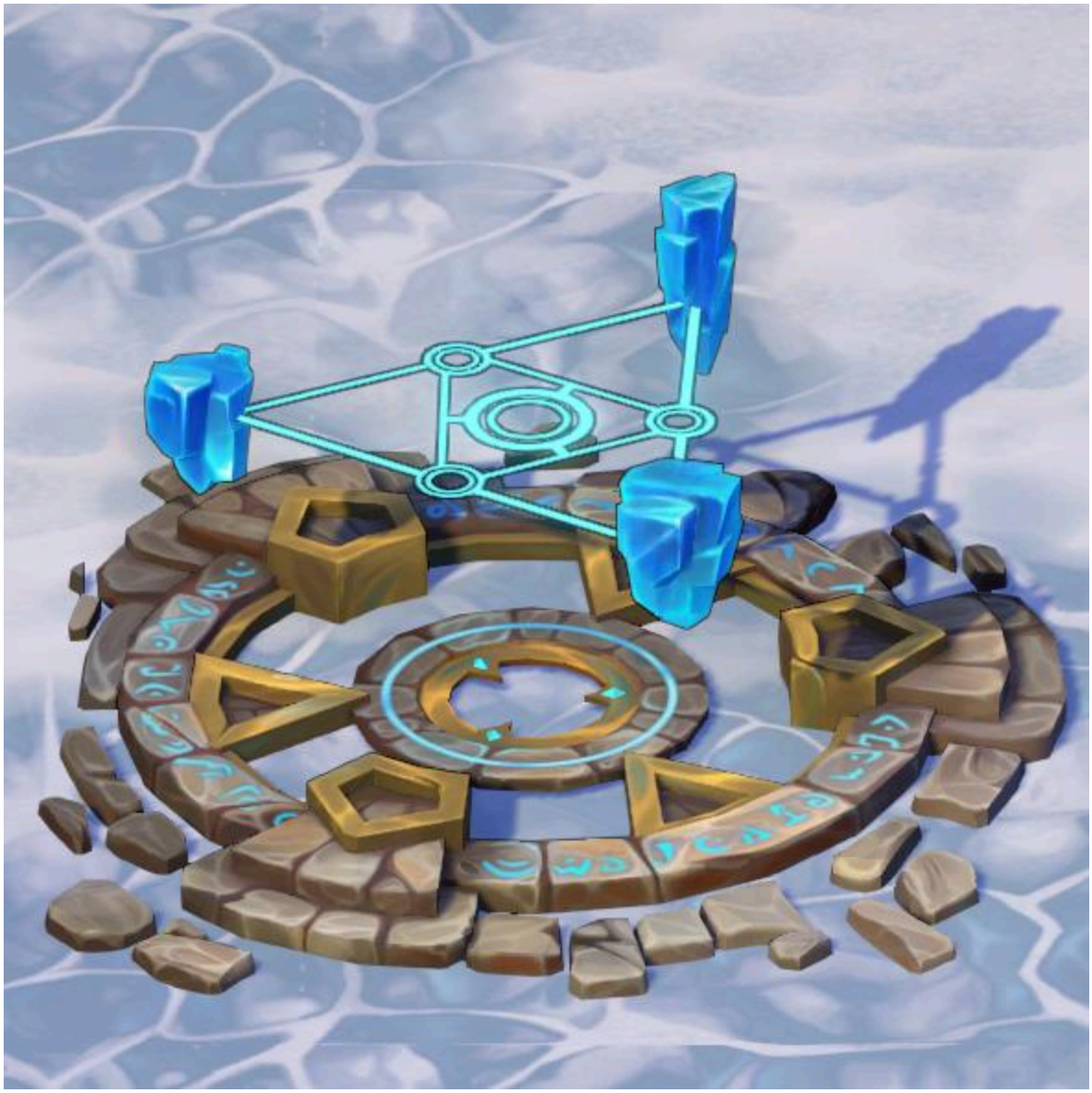
The player counts objects to determine how many are present or needed.



Stage B — Operations

Mathematical focus: operating on quantities.

The player constructs an expression involving quantities and an appropriate operation to resolve a resource or environmental puzzle.



Stage C — Relationships

Mathematical focus: representing relationships between quantities.

The player uses a mathematical relationship to determine how a mechanism should respond to changing inputs.



Stage D — Models and functions

Mathematical focus: expressing and investigating more complex relationships.

The player constructs or interprets a mathematical model to predict a phenomenon or control a more complicated magical interaction.

These are conceptual examples, not fixed stage names, spell tiers, or a prescribed sequence of curriculum competencies.

The underlying architecture remains:

Envision → Construct a mathematical expression → Command → Sprite interpretation → Response → Discovery.

What changes is the mathematical knowledge needed to construct and interpret the expression, as well as the complexity of the encounter.

IX. Educational progression and player mastery

20. Mathematical mastery should change what players can do

Status: Proposed design principle

Progression should not be limited to unlocking more sigils.

A player may become more capable through several forms of learning:

- Recognition: Identifying the relevant mathematical concept.
- Construction: Forming a valid mathematical expression.
- Application: Using the expression to solve a concrete problem.
- Prediction: Anticipating the consequences of a mathematical relationship.
- Interpretation: Understanding the results of an interaction.
- Transfer: Applying a learned concept in a different context.
- Adaptation: Revising an approach when the circumstances change.

These are possible dimensions of mastery, not finalized player statistics.

Example: Two players, the same counting sigil

Both players have access to the same counting-related sigil.

Player A can count a small collection when directly prompted but struggles when objects are distributed across the environment.

Player B can count the collection, record the quantity, check it against another representation, and apply the result to a new situation.

Both have access to the same magical component, but their demonstrated mathematical mastery differs.

The game can reflect this difference through the situations each player can solve independently, the accuracy of their work, and their ability to apply counting in unfamiliar contexts.

There is no need to invent two different counting sigils simply to represent two different levels of proficiency.

21. Avoid making the system a question-and-answer machine

The game can still use direct mathematical questions when they are appropriate. The problem is making them the only way mathematics appears.

A stronger encounter may ask the player to construct an expression, choose the appropriate mathematical representation, identify an error, predict an outcome, or investigate a pattern.

The best interaction depends on the competency being taught.

For instance, a counting encounter should provide an opportunity to count. A competency about representing a relationship should provide an opportunity to represent that relationship. A question asking for a final answer alone may not sufficiently assess either one.

This principle should guide the design of every future encounter.

X. Rules and boundaries of the magic system

The following rules consolidate the current direction into a practical reference.

Rule	Design requirement
Mathematical precision	Established sigils retain their exact mathematical meanings.
No arbitrary sigil meanings	A sigil cannot acquire an unrelated mathematical meaning merely to support a plot point.

Composition follows rules	Multiple sigils form meaningful expressions only when their arrangement is valid under the established language.
Mathematics is not fuel	Mathematical knowledge does not function as a personal reserve of magical energy.
Sprites are not batteries	Sprites are living manifestations with their own natures and capabilities.
Devices are interfaces	Magical devices facilitate communication and interaction; they do not replace understanding.
Reality imposes limits	Envisioning an outcome does not guarantee that it can be achieved.
Failure has causes	Mathematical errors, communication problems, Sprite limitations, and environmental conditions should remain distinguishable.
Mastery extends beyond answers	Mathematical competence should be demonstrated through meaningful application, not merely correct-answer entry.
Not every competency is a spell	Mathematical learning may also appear through exploration, investigation, decisions, and other RPG activities.
Fantasy retains mystery	Mathematical models can explain or predict aspects of a phenomenon without eliminating its supernatural nature.
Discovery remains central	The player should have opportunities to investigate, experiment, and learn more about the world.

XI. Open decisions and provisional elements

This version intentionally does not finalize every part of the magic system. The following matters still need development.

1. The formal grammar of sigils

We need to define how sigils are arranged, how values enter expressions, and how mathematical structures become valid spells.

2. The exact role of commands

The Two-Table system establishes Command as formal communication. The revised idea establishes sigils as components used to construct spells. The precise relationship between a command, its sigils, its inputs, and the resulting spell remains to be defined.

3. The representation of mathematical inputs

Values, variables, operations, relations, functions, and other mathematical structures may need different forms of representation. Their exact treatment is not yet locked.

4. Spell levels and classifications

The concepts of spell complexity, spell classification, and spell type should remain separate. Their final labels and mechanics have not been established here.

5. Sprite interpretation mechanics

The exact relationship between Sprite understanding, affinity, environmental conditions, and spell reliability remains open.

6. The relationship with Envision and Command

The new spell-construction model must be integrated into the existing Two-Table system without replacing its foundational distinction.

The next step should be to prototype the counting encounter and test the proposed interaction from beginning to end. Only after the basic model works should it be generalized to more complicated mathematical competencies.

XII. Final synthesis

The magic system can be understood as four interacting layers.

Layer 1 — Mathematical Language

Mathematical concepts, sigils, values, expressions, and the rules governing their construction.

Layer 2 — Magical Communication

Envision, Command, spell construction, and the interfaces through which intentions are expressed.

Layer 3 — Magical Interpretation

Magic Sprites interpret the communicated expression according to their nature, understanding, and capabilities.

Layer 4 — World and Response

Environmental conditions, magical phenomena, outcomes, feedback, and opportunities for further discovery.

These layers are a proposed organizational model, not a replacement for the established Two-Table architecture. They clarify the different responsibilities involved in spellcasting.

The educational progression emerges as players learn more mathematics, construct more sophisticated expressions, and discover new ways to interact with the world. The fantasy progression emerges as players encounter new Sprites, investigate unfamiliar phenomena, and learn the capabilities and limitations of magic.

The two progressions should support each other without becoming identical. Learning a mathematical concept should not automatically require a new Sprite, and meeting a new Sprite should not automatically require a new mathematical concept.

The central principle of MATHBOUND's magic system is this:

> Mathematics is not a magical power assigned to every concept. It is the precise language through which practitioners learn to express and interact with magical reality. Sigils make that language tangible, spells put it into use, and Magic Sprites bring the resulting interactions to life.

This gives MATHBOUND a scalable foundation for integrating mathematics into an RPG while preserving the world's mystery, the importance of discovery, and the educational purpose of the game.